Realtime U-Net-inspired Video Super Resolution

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Abstract

This research aims to address the challenge of enhancing low-resolution video content through the investigation of video super resolution techniques. We propose a novel U-Net based architecture that leverages information from previous frames to effectively process video frames at multiple resolutions. Our approach prioritizes computational efficiency, making it suitable for mobile devices, while aiming to achieve real-time performance for video streaming. We will train our models on a set of various nature-themed videos and evaluate its performance using quantitative measures such as Peak Signal-to-Noise Ratio (PSNR) and the Structure Similarity Index Metric (SSIM). Our goal is to reduce bandwidth usage without compromising video quality, with applications in domains such as video streaming platforms.

1. Introduction

In recent years, the proliferation of digital content consumption, particularly in the realm of video streaming, has accentuated the need for efficient methods to enhance the quality of low-resolution video content. This has applications in many industries such as medical, security, agricultural, entertainment, and manufacturing. While some approaches apply image super resolution models frame-by-frame, the additional video context can make predictions more accurate.

While convolutional neural networks (CNN) [7] and recently transformers [2] have been applied to the video super resolution task, our approach uses the U-Net architecture [9]. The U-Net, originally introduced for biomedical image segmentation, offers promising inductive biases for video super resolution. Unlike vanilla convolutional neural networks, which tend to compress images into semantic labels, U-Net emphasizes pixel-level prediction, making it well-suited for generating fine-grained detail in super resolved videos. Furthermore, U-Net is built upon efficient convolutional layers, aligning with our goal of achieving computational efficiency on mobile devices. We aim to decrease the latency and increase accuracy across PSNR and SSIM metrics for the video super resolution task.

2. Related Work

The foundation of video super resolution lies in convolutional neural networks (CNNs), which have been extensively optimized to discover optimal parameters for enhancing image quality. Early approaches focused on leveraging convolutional layers to capture spatial information effectively [7]. Subsequently, researchers scaled up these models by incorporating 3D convolution layers to better fuse spatial and temporal information, resulting in significant advancements in the field [1]. To create better prediction objectives that match human perception rather than naively weighting each pixel difference equally, works such as [6] innovate by introducing a discriminator model to evaluate predictions. Inspired by the success of transformers in language modeling tasks, recent research has explored their application to video super resolution [2]. However, transformers are known for their large memory usage, making them less suitable for deployment on resource-constrained devices such as mobile phones. We aim to leverage a novel U-net architecture and its ability to capture multi-resolution features for the video super resolution task.

3. Methods

3.1. Dataset

For this work, we use a dataset of copyright-free ocean-themed nature documentaries that we download from YouTube [3]. We down-sample the original 720p videos and use this lower resolution video as input to the model with the target being the original high-resolution video.

We selected this dataset as the ocean dataset was likely to contain more constrained motion patterns associated with marine animal motion, which would then be easier for our model to learn during the training phase.

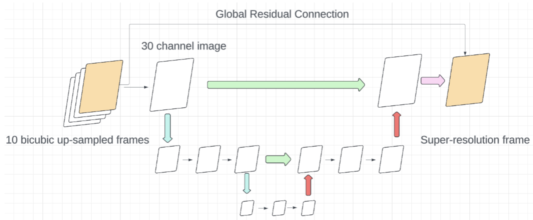

Figure 1. Overview of our novel Temporally-Augmented Residual U-Net Architecture for Video Super Resolution. The U-Net takes in a tensor composed of the last 10 frames concatenated along the color channels creating a 30-channel input to the U-net. The U-net then predicts the high-resolution frame.

For every frame, we include the 10 most recent frames up to the current one as input to our model to increase the amount of context for making the prediction.

3.2. Model

We approach our task of predicting each high-resolution video frame based on low-resolution video frames by first upsampling the low-resolution frames using bilinear upsampling and then using our network to improve the upsampled frame. We adapt the U-Net architecture introduced in [9] for image segmentation. We concatenate the frame tensors along the channel dimension as input and feed that into our U-Net which is modified to take a higher number of channel dimensions than three for the traditional RGB image. Our U-Net then outputs an image with three channels. The architecture is depicted in 1.

3.3. Training

We train on a RTX 4090. We consider two tasks, one where the upsampling/downsampling ratio is 4, used in an oft-cited paper [6], and the second where the ratio is 16. We perform numerous experiments, and for each we train for 5000 steps, checkpoint every 1000 steps, take the best checkpoint on our test set, and utilize AdamW [4] optimizer with a learning rate of 1E-4 and $\beta_1 = 0.9, \beta_2 = 0.999$. We train two types of models on different resolution images. The target resolution is 720p. For one type of models, we train it to predict frames from 180 or 45p frames. For the other, we predict 80p square patches from 20 or 5p square patches and tile all the predictions to form the target prediction. For loss, we use average mean-squared error (MSE) loss over the pixels. We also experiment with a loss given by one minus the structural similarity index measure (SSIM), which attempts to measure reconstruction quality as perceived by humans, on the full-sized predictions [10]. We also provide the code used for all training and evaluation in the following repository for reproducibility: https://github.com/hdeep03/U-Net-VSR.

3.4. Evaluation

We evaluate the model on the validation and test sets using Peak Signal to Noise Ratio (PSNR) [5] which is an absolute measure of the reconstruction error and SSIM [10]. For our test set, we select 5 diverse images from our ocean videos and compare model outputs visually and numerically. We also evaluate efficiency of our model on the Apple macbook pro with M1 Pro processor to see if our model can be used in real-world use cases on consumer hardware.

4. Results and Discussion

Our best-performing models are the residual U-Net architecture trained with MSE loss predicting patches when we are downsampling/upsampling by 4, and without patches and SSIM loss when the ratio is 16. Model outputs on the test set are shown in 5 and the below tables, with the model-generated frame on the left and low-resolution upsampled frame on the right. Also, we calculate the PSNR and SSIM improvements of our models over the low-resolution baseline. We vary model size, the loss function we use, and whether or not we use patches, which are the three most promising parameters to change given our experiments. We display results in Table 1. We also train an Efficient Sub-Pixel Convolutional Neural Network (ESPCN) [1] on our dataset with the same compute budget of 5000 steps, where each step the ESPCN model generates a 16×9 patch-grid just like how we train models with patches. We choose this model to compare with because it highly efficient. We find a significantly worse PSNR and visual inspection indicates the ESPCN model still contains notable artifacts. This suggests our Residual U-Net architecture has superior inductive bias.

4.1. Efficiency Analysis

We profile our code and find that we can utilize our model real-time if we have a GPU. We find slight differences between each architecture, each taking around 0.0027 seconds per image and 2.55GB GPU memory. We believe that the highly convolutional nature of our architecture makes inference efficient especially on modern GPU accelerators. We further test our model on our Apple M1 laptops, achieving frame latencies depicted in Table 1. Our model is slower than ESPCN by two orders of magnitude but performs better in terms of reconstruction loss evinced by the higher PSNR and SSIM.

4.2. Global Residual Connection

We find that augmenting a U-Net architecture with a global residual connection greatly improves the performance of our model. When performing an ablation experiment on a Residual U-Net architecture trained with MSE loss and with patches, we find that changing the architecture to normal U-Net leads to an over 10 dB drop in PSNR and 0.03 drop in SSIM. Thus we enable the global residual connection by default.

4.3. Down-scaling model size

We experiment with utilizing a smaller U-Net model with 2 layers of upsampling and downsampling rather than 4. This decreases the parameter count from 31.1 million to 1.8 million and single-image inference time from 0.0027 to 0.0015. We observe that the model training loss is definitely smaller for a larger model, yet the validation loss for both does not have a clear conclusion. Our resources are constrained and since the model is quite undertrained, only getting through 0.01 of an epoch, with more training, we believe the superiority of the larger model would manifest. At such early stages, the superior train loss in the larger model is more meaningful than the worse validation loss. Indeed, we see in 1 that the small model's performance drops, except notably with SSIM loss and no patches the small model achieves similar performance to the larger one.

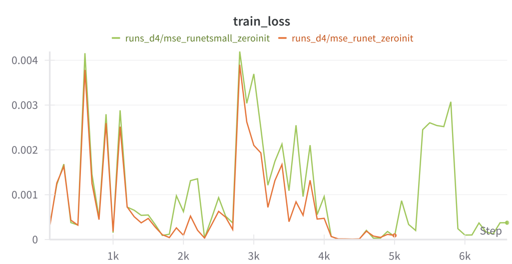

Figure 2. Training curve for the smaller and larger model shown in green and orange respectively, illustrating lower loss and better reconstructions from the larger model.

4.4. Image vs. Video prediction

We do not find a convincing advantage for using 10 images as context versus using just the current low-resolution image. Our experiment compares the models with downsampling ratio of 4 and without predicting patches. In 3, we see sometimes the validation loss is lower and sometimes it is higher. On the test set, in fact using one image as context for the large model with MSE training loss and no matches achieves superior PSNR and SSIM differences with respect to the low-resolution upsampling baseline method of -0.12 and -0.00076, compared to -0.57 and -0.0020 of the counterpart using 10 images. This is contrary to our intuition of having more temporal context helping, and it may have to do with the increased model complexity of having more contextual images than one making it harder to learn with limited compute.

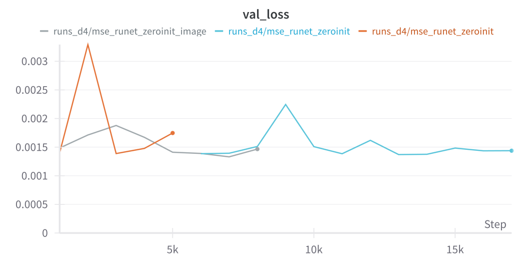

Figure 3. Validation MSE loss on number of contextual images

4.5. Tanh instead of ReLU activation functions

When x is small and positive, $\mathrm{Tanh}(x) \equiv \frac{e^x - e^{-x}}{e^x + e^{-x}} \approx (2x)/2 = x = \mathrm{ReLU}(x)$. When we zero-initialize our residual U-Nets and train them, it appears that the weights converge to small positive numbers, causing the near-exact overlap in training dynamics between using ReLU and Tanh activation functions throughout our model in 4.

Figure 4. Train MSE loss

5. Conclusion

We have explored the space of U-Net-based architectures for Video super resolution. We have built Residual U-Nets that marginally achieve superior to naive upsampling performance and can be run in real-time. Our models are undertrained, as evidenced by the poor performance of the successful ESPCN model with the same amount of compute, and further work can validate our conclusions when trained with more steps. Remaining research directions include refining the objective function as some weighted combination of MSE and SSIM loss, or branching out to other types of losses such as perceptual loss as explored in [8]. Moreover, the direction of utilizing past frames to improve predictions of the current frame could be further explored by creating architectures involving attention or recurrence to capture more complex relationships and weight more relevant frames higher than others.

Model	Patches	Training Loss	Test PSNR - LR PSNR	Test SSIM - LR SSIM	Frame Latency (ms)
ESPCN	✓	MSE	-11.58	-0.049	2.72 ± 0.21
Small	✓	MSE	-0.28	$-1.93 \cdot 10^{-6}$	227 ± 11
Small	✓	SSIM	-0.13	$-1.31 \cdot 10^{-7}$	238 ± 24
Small	×	MSE	-0.41	$-1.3 \cdot 10^{-5}$	224 ± 11
Small	×	SSIM	1.15	-0.00015	273 ± 67
Large	✓	MSE	0.60	$3.1 \cdot 10^{-6}$	527 ± 167
Large	✓	SSIM	0.95	$-3.6 \cdot 10^{-5}$	491 ± 97
Large	×	MSE	-0.57	-0.0020	420 ± 72
Large	×	SSIM	1.15	-0.00015	414 ± 57

Table 1. Comparison of Model Performance. Frame latency is the time to generate one frame using each model.

6. Author Contributions

Harsh worked on compiling and preprocessing the dataset, writing the dataloaders, benchmarking the latency of all the models, and running the experiments with patch models that only operated on a patch of the image. Alex ran the experiments for testing the Tanh vs ReLU activation function, an ablation study for the global residual connection, and experiments testing whether there was a significant different in performance using 1-frame vs 10-frames. Rithvik worked on writing the evaluation scripts, creating the test image set, and running the experiment comparing model performance when using the MSE and SSIM loss functions.

References

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Figure 5. From left to right, model output, high-resolution original, low-resolution bicubic upsample (same applies); Model (MSE, patches, ratio 4): 50.60 dB / 0.0036, Bicubic: 49.24 dB / 0.0036

Figure 6. From left to right, model output, high-resolution original, low-resolution bicubic upsample; Model (MSE, patches, ratio 4): 35.97 dB / 0.03783, Bicubic: 35.95 dB / 0.03785

Figure 7. From left to right, model output, high-resolution original, low-resolution bicubic upsample; Model (MSE, no patches, ratio 16): 26.64 dB / 0.1492, Bicubic: 26.55 dB / 0.1515

Figure 8. From left to right, model output, high-resolution original, low-resolution bicubic upsample; Model (MSE, no patches, ratio 16): 23.54 dB / 0.1220, Bicubic: 23.49 dB / 0.079